System Flowchart

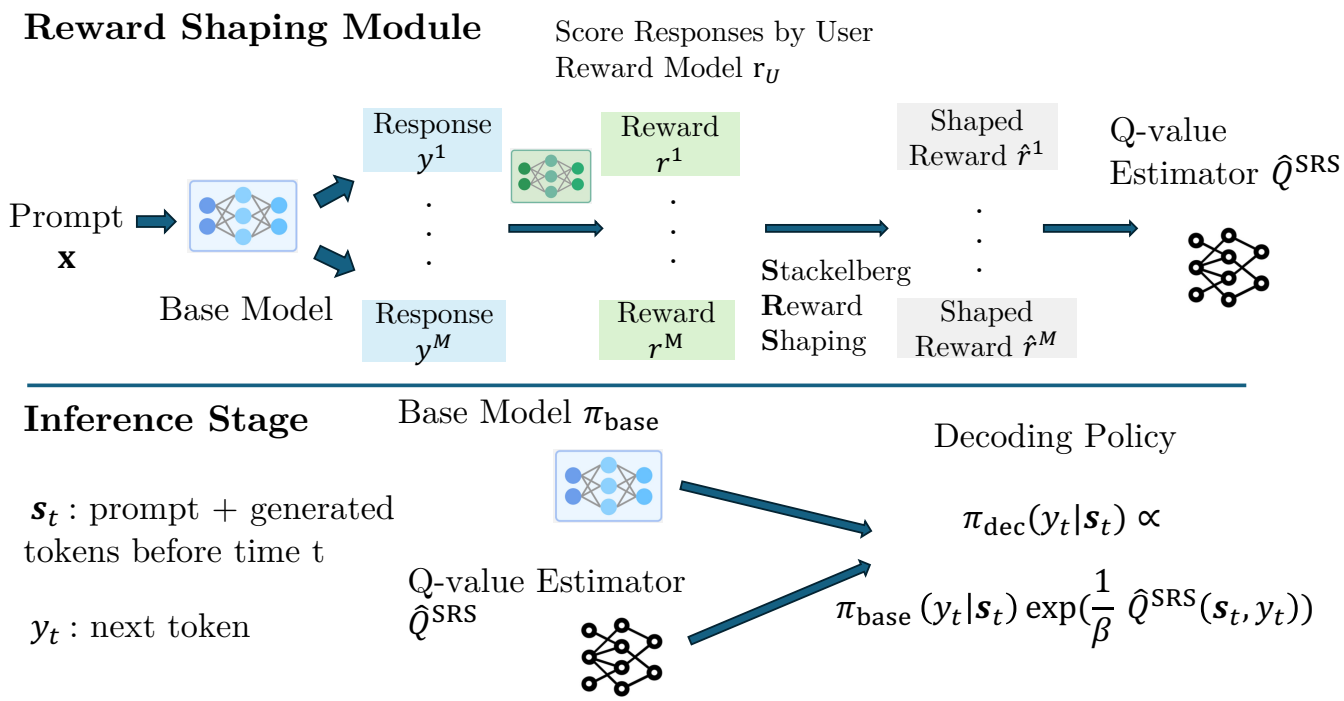

Figure 1: Overall framework of Stackelberg Reward Shaping. We first sample multiple responses from the base model and evaluate them using the user reward model, followed by Stackelberg reward shaping to maximize user utility. This process produces a Q-function estimate $\hat{Q}^{\mathrm{SRS}}$, which captures the expected reward of completed responses under the shaped reward. During inference, $\hat{Q}^{\mathrm{SRS}}$ is used to reweight the base model's token probabilities, promoting tokens aligned with user preferences.